Mixed-Precision MAC Unit

Design of Multiplier-Accumulator with Different Operand Widths and Wider Accumulator
Basics, RTL Design, Simulation and Exercises

Agenda

1. Introduction
2. Need for Mixed Precision
3. MAC Architecture
4. Verilog RTL
5. Testbench
6. Simulation Results
7. Exercises & Solutions

What is a MAC Unit?

MAC performs: Accumulator = Accumulator + (A × B).
Used in DSP, AI accelerators, filters, image processing and ML inference.

Mixed Precision Concept

Operands can have different widths. Example:
A = 8-bit
B = 16-bit
Product = 24-bit
Accumulator = 32-bit or 40-bit
Benefits: lower area, power and higher flexibility.

Architecture

8-bit Operand A --> Multiplier <-- 16-bit Operand B
Product Width = 24 bits
Wider Accumulator = 40 bits
ACC <= ACC + PRODUCT

RTL Verilog Code

```
module mixed_precision_mac #(parameter AW=8,BW=16,ACCW=40)(
input clk,rst,en,
input [AW-1:0] a,
input [BW-1:0] b,
output reg [ACCW-1:0] acc);
wire [AW+BW-1:0] product;
assign product=a*b;
always@(posedge clk) begin
 if(rst) acc<=0;
 else if(en) acc<=acc+product;
end
endmodule
```

RTL Explanation

AW = operand A width
BW = operand B width
ACCW = accumulator width
Product width = AW + BW
Accumulator wider than product prevents overflow.

Testbench

```
module tb;
reg clk=0,rst,en;
reg [7:0] a; reg [15:0] b;
wire [39:0] acc;
mixed_precision_mac dut(...);
always #5 clk=~clk;
initial begin
rst=1; #10 rst=0; en=1;
a=8'd10; b=16'd20;
#10; a=8'd5; b=16'd30;
#20; $finish; end
endmodule
```

Expected Simulation

Cycle1: Product=10×20=200, ACC=200
Cycle2: Product=5×30=150, ACC=350
Waveforms should show accumulator increasing every enabled clock edge.

Exercise 1

Design a MAC with 16-bit × 16-bit operands and 48-bit accumulator.
Find product width and write RTL.

Exercise 1 Solution

Product width = 32 bits.
Accumulator = 48 bits.
Use parameterized RTL with AW=16, BW=16, ACCW=48.

Exercise 2 & Summary

Exercise: Add saturation logic when accumulator overflows.
Exercise: Support signed multiplication.
Summary: Mixed-precision MAC balances performance, area and power while maintaining accuracy through a wider accumulator.